

eDrishti

Project Introduction and History:

Our project focuses on overcoming the educational challenges visually impaired individuals encounter in science and mathematics. We discovered that globally only 5% of visually impaired people are literate, with just 1% in India. The lack of Braille study materials and tactile training in government schools worsens the situation. After learning about the struggles of visually impaired students through legal cases and personal conversations, we identified the need for a solution. Our primary goal is to make science and mathematics education accessible to visually impaired individuals beyond the 8th standard.

Proposed Solution:

To address this need, we propose a software-embedded system consisting of a mobile application and a connected hardware system. The application employs advanced voice explanation and visualization techniques to help visually impaired users understand STEM concepts. A tutor guides the creation of learning modules, ensuring relevance and accuracy. The application supports TalkBack and voice control features for ease of use. The hardware involves pins that rise and fall based on geometry traced on the software, enabling users to touch and understand the figures.

Team Introduction:

Team E-DRISHTI, comprising undergrads Ishita Kapoor and Adarsh Goyal, aims to impact the education industry. Ishita, a final year undergrad and software developer, and Adarsh, a pre-final year student excelling in hardware development, share a passion for addressing educational inequalities. Their diverse skills contribute to the development of an innovative solution for visually impaired individuals.

Motivation for the Project, Inspiration:

The project was inspired by the stories of visually impaired individuals, particularly Dhaninjay, encountered during visits to the National Blind Institution. Witnessing the struggles these individuals faced due to limited access to STEM education motivated the team to create an affordable tool that could bridge the educational gap and empower visually impaired students.

The Need - Why This is Important:

Our project addresses the significant educational challenges faced by visually impaired individuals in STEM education. It eliminates the need for physical assistance, hefty tutoring fees, or expensive braille books. The proposed system aims to increase employability and literacy rates among visually impaired individuals in the target country.

Technology Used:

The core of our solution combines hardware and software. Flutter was chosen for software development, integrating Azure services like CosmosDB, API Management, Notifier, and CI/CD pipelines. Hardware components include Arduino UNO, Solenoid Actuators, and an HC-05 Bluetooth Module. This combination enables users to interact with digital representations of scientific and mathematical concepts.

Impact it Can Make or Has Made:

The prototype received positive feedback during user testing, leading to improvements in both hardware and software. Collaborations with student welfare organizations, including a UNICEF subsidiary, demonstrate our commitment to positive community impact. With successful testing by over 100 visually impaired individuals, the final version is expected to be a success.

Plans to Expand:

Future plans involve refining the prototype, expanding outreach through partnerships with NGOs and blind schools, and leveraging social media. Sample pieces for practice will be distributed to build a substantial user base before the official product launch. Exploring collaborations with organizations like Open AI is also on the agenda to enhance technological capabilities.

Experience with Users:

User feedback, especially from visually impaired individuals, has played a crucial role in refining the project. The positive response, pre-orders, and expressions of gratitude motivate the team to continue refining and expanding the project, ensuring it meets the specific needs of the target audience.

In conclusion, Team E-DRISHTI is committed to addressing educational disparities for visually impaired individuals in STEM education. Through innovative technology and collaborations, the project aims to empower and uplift the visually impaired community, providing equal opportunities for academic and career pursuits.